Electrical Engineering Department

RapidLaneFGPA: FPGA DPU-Based Highway Lane Segmentation System

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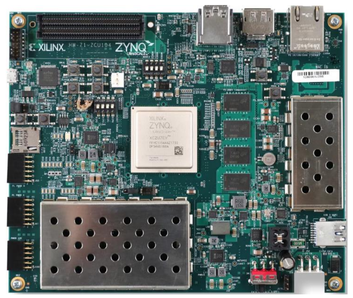
Introduction

RapidLaneFPGA provides a rapid iteration of designing, training, quantizing, and deploying a highway lane segmentation system using a neural network. Advanced Driver Autonomous Systems (ADAS) engineers need a practical way to deploy custom neural networks to custom hardware, and this project explores leveraging different software and hardware workflows to make software neural networks realizable on targeted reconfigurable hardware.

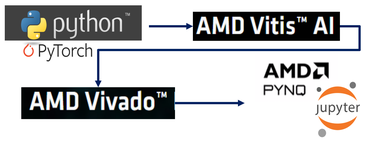
Traditionally, once neural networks are conceptualized within a software integrated development environment, there is not a trivial clear path to run inference on ***standalone*** hardware such as a microprocessor, microcontroller, or Field Programmable Gate Array (FPGA). RapidLaneFPGA aims to rapidly deploy a practical lane segmentation system on a AMD ZCU104 Ultrascale+ FPGA.



Methodology

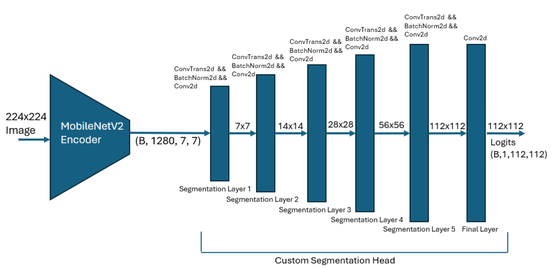
RapidLaneFPGA implements a software – hardware codesign that leverages software libraries and hardware workflows to run inference on custom hardware.

A.) Design an implement a lane segmentation model using a pretrained MobileNetV2 encoder and a custom decoder head
B.) Train and fine tune the model with 224x224 pixel resolution using a CUDA enabled GPU to yield the best model
C.) Export the weights and biases from the trained model for quantization using AMD's Vitis AI workflow to produce a hardware .XMODEL
D.) Design and synthesize an RTL design leveraging ZYNQ processor, Deep Processing Unit, and support infrastructure
E.) Load bitstream and design files into AMD PYNQ framework to run pure inference on the FPGA fabric

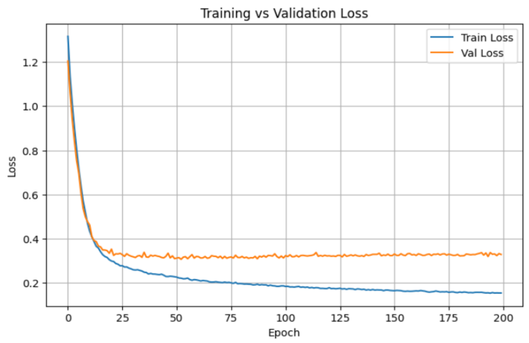


Methodology

A.) The model is built using base encoder from the MobileNetV2 neural network for its' powerful feature extractor, lightweight parameter load, and hardware friendly implementation. The model deploys a custom segmentation head that up samples the encoder output to spatial resolutions in the order of 7×7 → 14×14 → 28×28 → 56x56 → 112×112.

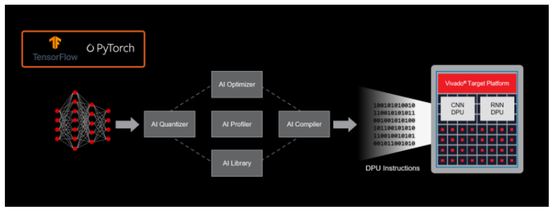


B.) Training the model included finetuning hyper-parameters such as learning rates, epochs, loss criterions, weight decays, transformations and augmentations.

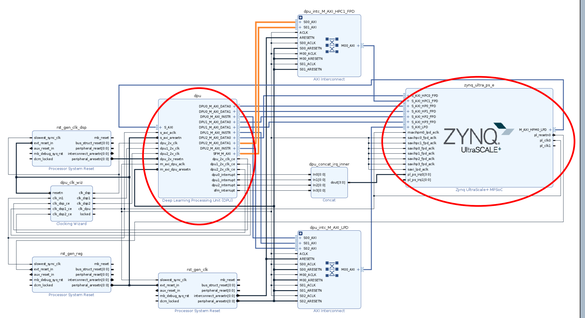


Loss Function	Batch Size	Learning Rate & Weight Decay	Epochs	Training Time	Dice Score Train	Dice Score Test	Transformation
BCE+DiceLoss (Best)	16	1×10^{-4} 1×10^{-4}	200	3.3 Hours	88.69%	77.81%	Yes
BCEWithLogitsLoss	16	1×10^{-4} 4×10^{-4}	200	3.3 Hours	N/A	74.16	No
BCE+DiceLoss	16	5×10^{-4} 1×10^{-4}	200	3.5 Hours	83.27%	77.60%	Yes
BCE+DiceLoss	16	5×10^{-5} 1×10^{-4}	225	3.6 Hours	83.98%	77.10%	Yes
BCEWithLogitsLoss	16	1×10^{-3} 1×10^{-4}	200	3.3 Hours	93.94%	74.72%	Yes

C.) RapidLaneFPGA exports weights and biases from software to leverage VITIS AI workflow quantization into INT8 representation and compilation to produce XMODEL that runs inference on a Deep Processing Unit.



D.) The overall hardware design is synthesized using AMD Vivado block diagram tool that instantiates a ZYNQ processor, Deep Processing Unit, and support. infrastructure.



E.) AMD's PYNQ SoC framework is used to interface with the ZYNQ processing system and FPGA to command the neural network, load and offload data from DDR.

AMD PYNQ

Analysis and Results

The DICE coefficient algorithm is the main metric to test accuracy and performance of the model. The algorithm measures the magnitude of overlap of the predicted mask in comparison with the ground truth. If the coefficient is 1.0 (100%), that represents a full overlap where every pixel in the prediction matches every pixel in the ground truth.

$$\text{Dice Coefficient} = \frac{2 \times |A \cap B|}{|A| + |B|}$$

- Software DICE accuracy yielded 77.81% on the test dataset (363 images)
- Hardware DICE accuracy yielded 77.18% on the test dataset (363 images) at 10 Frames Per Second (FPS)

Software Overlay Output

Hardware Overlay Output

Implementation	Frames Per Second (FPS) (363 Images Test)
Software (Nvidia GeForce RTX 3060ti GPU)	~1.57 FPS
Hardware (ZCU104 Ultrascale+)	~10 FPS

Summary/Conclusions

RapidLaneFPGA yielded an acceptable DICE accuracy score within the software implementation.

The hardware implementation successfully produced near software DICE accuracy score which is expected. The FPGA performs inference at near real time speeds marking the project a viable proof of concept.

The entire software – hardware codesign is validated from the performance and results from the project.

Key References

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